



Tuning and Testing

Tuning and Testing involves the following:

1. Iterative tuning methods,
2. Validation through testing, and
3. Implementing safety measures



Alt text: A person in testing lab with a drone

Iterative Tuning Methods

Tuning the PID controller process involves adjusting the proportional(P), integral(I), and derivative (D) gains iteratively, testing different values, and observing their effects to achieve a balance where the drone responds quickly and accurately to control inputs without excessive oscillations or delays. This process is typically iterative, meaning it requires multiple rounds of adjustments and testing to find the optimal settings.

Example:

1. Start with Proportional Gain (K_p):

- Begin by setting the integral and derivative gains to zero.
- Increase the proportional gain incrementally until the drone responds quickly to control inputs. The goal is to achieve a fast response without causing the drone to oscillate around the setpoint.
- If the drone starts to oscillate (move back and forth around the setpoint), reduce the proportional gain slightly until the oscillations are minimised.

2. Add Integral Gain (K_i):

- With the proportional gain set, gradually introduce the integral gain.
- The integral gain addresses any steady-state error (the drone not maintaining the set altitude over time). Increase it until the drone maintains the set altitude accurately without drifting.
- If the integral gain is too high, the drone may start to oscillate again. Find a balance where the drone stays at the correct altitude without persistent oscillations.

3. Introduce Derivative Gain (K_d):

- Finally, add the derivative gain to dampen any remaining oscillations and prevent overshoot.
- The derivative gain reacts to the rate of change of the error, smoothing out abrupt movements. Increase it gradually until the drone's altitude changes are smooth and controlled.
- If the derivative gain is too high, it can cause the control response to be sluggish. Adjust it to achieve a balance between smoothness and responsiveness.

Validation Through Testing



Testing the tuned PID controller in various conditions ensures it performs well in real-world scenarios. This involves conducting a series of test flights under different environmental conditions and operational modes to validate the effectiveness of the tuning.

Example:

1. Test in Different Weather Conditions:

- Conduct test flights in calm weather, windy conditions, and varying temperatures to observe how well the PID controller maintains stability and altitude.
- For example, in windy conditions, ensure the drone can resist wind gusts and maintain its flight path without excessive deviation.

2. Observe Stability and Response:

- Monitor how well the drone maintains altitude, handles directional changes, and responds to control inputs.
- For instance, if the drone is used for aerial photography, ensure it can hover steadily for clear shots and respond smoothly to position adjustments.

3. Adjust PID Gains as Necessary:

- Based on the test results, make fine adjustments to the PID gains to optimise performance. This might involve slightly increasing the derivative gain if the drone still oscillates in gusty conditions or tweaking the integral gain for better altitude maintenance.

Implementing Safety Measures

Safety measures are critical to prevent crashes and ensure the drone can handle unexpected situations. Implementing these features enhances reliability and ensures safe operation under various conditions.

Example:

1. Automatic Return-to-Home (RTH):

- Implement a feature that automatically returns the drone to its home location if it loses signal or if the battery level drops below a critical threshold.
- For example, if the drone loses connection with the controller, it should autonomously fly back to its starting point and land safely.

2. Set Safe Limits for Altitude and Speed:

- Define maximum altitude and speed limits to prevent the drone from entering dangerous flight conditions.
- For instance, set a maximum altitude of 120 meters to avoid restricted airspace and a maximum speed of 20 m/s to ensure stable flight.

3. Emergency Procedures:

- Program emergency procedures for situations such as motor failure or GPS signal loss. These could include safe landing protocols or hovering in place until control is regained.
- For example, if a motor fails, the drone should execute a controlled descent using the remaining motors.

Practical Steps for Tuning and Testing

Initial Setup:

Set up a safe testing environment where the drone can be flown without risk to people or property. Ensure the drone is equipped with all necessary safety features, such as propeller guards and a GPS module for location tracking.

Example:

- Use an open field with minimal obstacles for initial test flights.
- Equip the drone with safety features like propeller guards, a GPS module, and a failsafe system to prevent accidents.

Proportional Gain Tuning:

Start with the proportional gain and gradually increase it until the drone begins to respond to control inputs with the desired speed but starts to oscillate.

Example:

- Set the integral and derivative gains to zero.
- Increase the proportional gain until the drone quickly corrects its altitude when disturbed.
- If the drone starts to oscillate, reduce the proportional gain slightly to find the optimal setting.

Integral Gain Tuning:

Introduce the integral gain to eliminate any steady-state error. Increase it gradually until the drone maintains its altitude accurately without drifting.

Example:

- Gradually increase the integral gain while monitoring the drone's ability to maintain a stable altitude.
- If the drone hovers slightly below the set altitude, increase the integral gain until it stays level at the desired height.

Derivative Gain Tuning:

Add the **derivative gain** to reduce overshoot and dampen oscillations. Increase it until the drone's movements are smooth and stable.

Example:

- Increase the derivative gain to smooth out any abrupt changes in altitude.
- If the drone's altitude changes are too abrupt and it overshoots, increase the derivative gain until the altitude changes are smooth and controlled.

Final Testing and Adjustment:

Test the drone in various conditions and fine-tune the PID gains as necessary to ensure reliable performance.

Example:

- Perform test flights in different weather conditions, during rapid manoeuvres, and with different payloads.
- Observe how the PID controller maintains stability and control across all scenarios, making fine adjustments to the gains as needed.